

Nilay Tripathi

Email: tripat55@msu.edu
Website: nilayt321.github.io

Location: East Lansing, MI, USA
GitHub: NilayT321

Areas of Interest:

Low dimensional topology, quantum topology, and knot theory.

Education:

- 09/25–Present **Michigan State University** *Doctor of Philosophy*
Second year doctoral student
- 09/21–05/25 **Rutgers University, New Brunswick** *Undergraduate Degree*
B.S. in Mathematics (high honors),
B.A. in Statistics (highest honors), and B.A. in Computer Science
summa cum laude, CGPA: 3.907

Talks & Presentations:

The following talks were delivered at Rutgers as an undergraduate.

- 05/25 “Integrable Systems in Hamiltonian Dynamics”
Independent study talk.
- 11/24 “Homology & Its Applications”
Undergraduate seminar talks by RUMA.
- 09/24 “Smooth Manifolds & Symplectic Geometry”
Rutgers Directed Reading Program
- 05/24 “de Rham Cohomology”
Class talk for Math 412.
- 12/23 “Banach & Hilbert Spaces: An Introduction to Functional Analysis”
Class talk for Math 441.
- 12/23 “An Overview of Hermite Polynomials”
Rutgers Directed Reading Program.

Mentorship Experience

- 05/26–06/26 **MSU Summer Topology Program**
Program mentor for undergraduates to learn a topic in geometry and topology.

Teaching Experience

At Michigan State University:

08/26–12/26 Recitation TA for MTH 234.
MLC Supervisor: oversee other tutors, students, and general center operations.

06/26–08/26 TA for MTH 124 (Survey of Calculus I)

01/26–05/26 MLC Tutor for MTH 132 (Calculus I)

09/25–12/25 Math Learning Center (MLC) Tutor for MTH 234 (Multivariable Calculus).

At Rutgers University (undergraduate):

01/25–05/25 Recitation LA for CS 112 (Data Structures)
Math Department Grader for Math 412 (Honors Mathematical Analysis II) and
Math 350 (Linear Algebra)

09/24–12/24 Recitation LA for CS 111 (Introduction to Computer Science)
Math Department Grader for Math 441 (Introductory Topology I) and Math 311
(Introductory Real Analysis)

09/23–05/24 Lecture LA for Math 300 (Introduction to Mathematical Reasoning)

09/22–05/23 Study Group Learning Assistant (LA) for Stat 212 (Statistics II)

Skills

- Languages: English (native), Hindi (native), French (beginner)
- Programming Languages: Python, C/C++, MATLAB, Java, $\text{\LaTeX}$, R
- Software: Git/GitHub, Linux (Ubuntu, Fedora/RHEL), Jupyter Notebook/JupyterLab